Mid 2022

Tags

1. a) Consider the network topology of a multi hop wireless network illustrated in Figure 1. In the given scenario, stations *A, B, C, D,* and *E* all have qui-sized transmission ranges which are depicted by circle surrounded by respective stations. Assume that all the stations operate in same frequency band and two stations transmissions will interfere if and only if they transmit at the same time and their transmission areas overlap. Consider the *RTS/CTS* transmission is a disabled in this scenario. 5.5 (CO1) (PO2)

   i. For the given scenario, identify four (4) possible cases of *Hidden Station Problem*. Your answer will provide the possible hidden stations for a pair of sender and receiver in a table.

   ii. Why does the *Hidden Station Problem* arise?

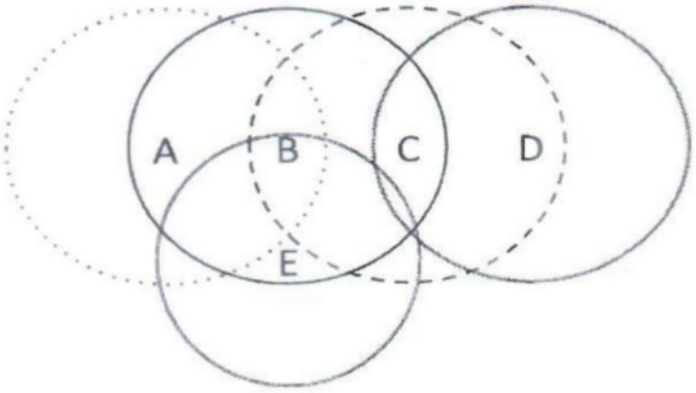
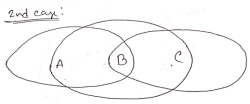
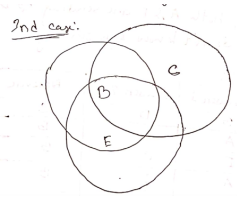


Figure 1: Network topology for question 1 (a)

i)







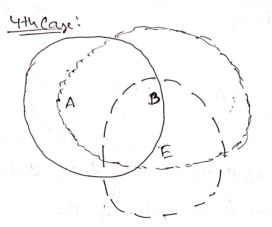
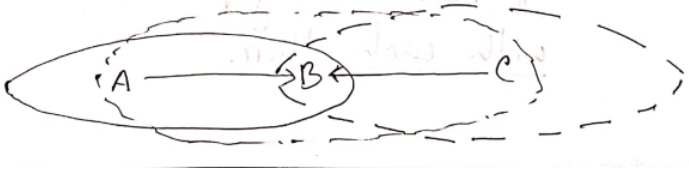


Table:

Sender 1	Sender 2	Reciever
B	D	C
A	C	B
E	C	B
A	E	B

ii)

When, one station (A) is sending to (B), (C) doesn't get any information of this transmission as, C is not in the range of A. Hence, C also sends to B &, collision occurs at B.

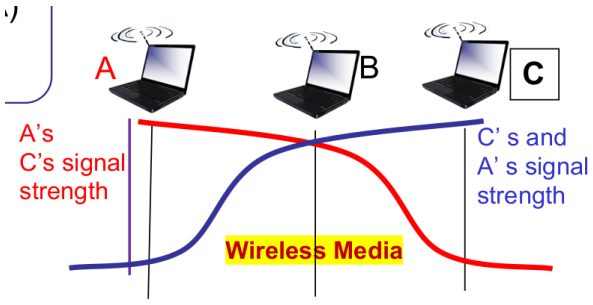


b) Analyze the limitation of detecting collision by a transmitting station in wireless network. Your answer should include appropriate justification and diagram. 4.5 (CO2) (PO2)

Signal State at Receiver (node B) and Transmitter (Node A) are NOT Identical. No Fate sharing of the link.

Collision experienced at Receiver (node B) while both node A and C are transmitting

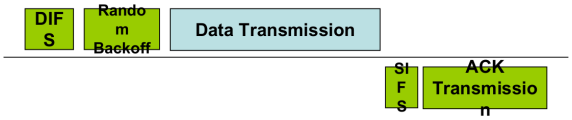
Collision Detection at Transmitter (Node A) is NOT feasible, by sensing(receiving) the signal strength available now by additional receiving antenna while transmission is going on



Hence, Cannot take advantage of detecting multiple transmission by simultaneously transmitting and receiving. We need Link Layer ACK to solve this.

Solution:

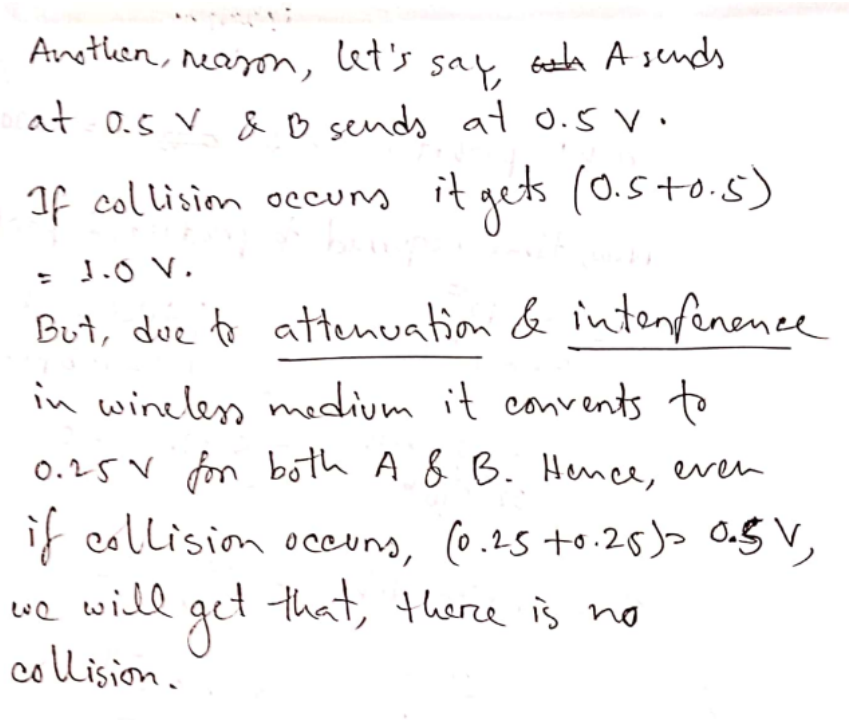
Sender interprets packet loss as being caused by collision



Problem: Does not handle hidden terminal cases.

Wireless technologies, in contrast to wired technologies, like IEEE 802.3 technologies, operate in half-duplex. It means that wireless devices (AP or an endpoint) can talk or listen however not simultaneously with a single antenna

- Attenuation, Inference and Noise causes problems


Another reason, let's say, ~~eda~~ A sends at 0.5 V & B sends at 0.5 V.
If collision occurs it gets (0.5+0.5)
= 1.0 V.
But, due to attenuation & interference in wireless medium it converts to 0.25 V for both A & B. Hence, even if collision occurs, (0.25+0.25)> 0.5 V, we will get that, there is no collision.

c) In this problem, we consider sending real-time voice from Host A to Host B over a packet-switched network (VoIP). Host A converts analog voice to a digital 32 mbps bit stream on the fly. Host A then groups the bits into 2-kilobyte packets. There is one link between Hosts A and B; its transmission rate is 20 Mbps and its propagation delay is 20 msec. As soon as Host A gathers a packet, it sends it to Host B. As soon as Host B receives an entire packet, it converts the packet's bits to an analog signal. 3.5 (CO3) (PO1)

How much time elapses from the time a bit is created (from the original analog signal at Host A) until the bit is decoded (as part of the analog signal at Host B)?

Consider the first bit in a packet. Before this bit can be transmitted, all of the bits in the packet must be generated. This requires

2 kB packet = 2×10^3 ~~bits~~ $\times 8 = 16000$ bits

Now, time required to produce 1 packet,

$= \frac{2 \times 10^3}{32 \times 10^6}$ | taking it as 32 Mbps.

$= \frac{16000 \text{ bits}}{32 \times 10^6} = 5 \times 10^{-4}$ sec

Transmission delay $= \frac{16000}{20 \times 10^6}$

$= 8 \times 10^{-4}$ sec

& propagation delay $= 20 \times 10^{-3}$ sec

So, total $= 5 \times 10^{-4} + 8 \times 10^{-4} + 20 \times 10^{-3}$

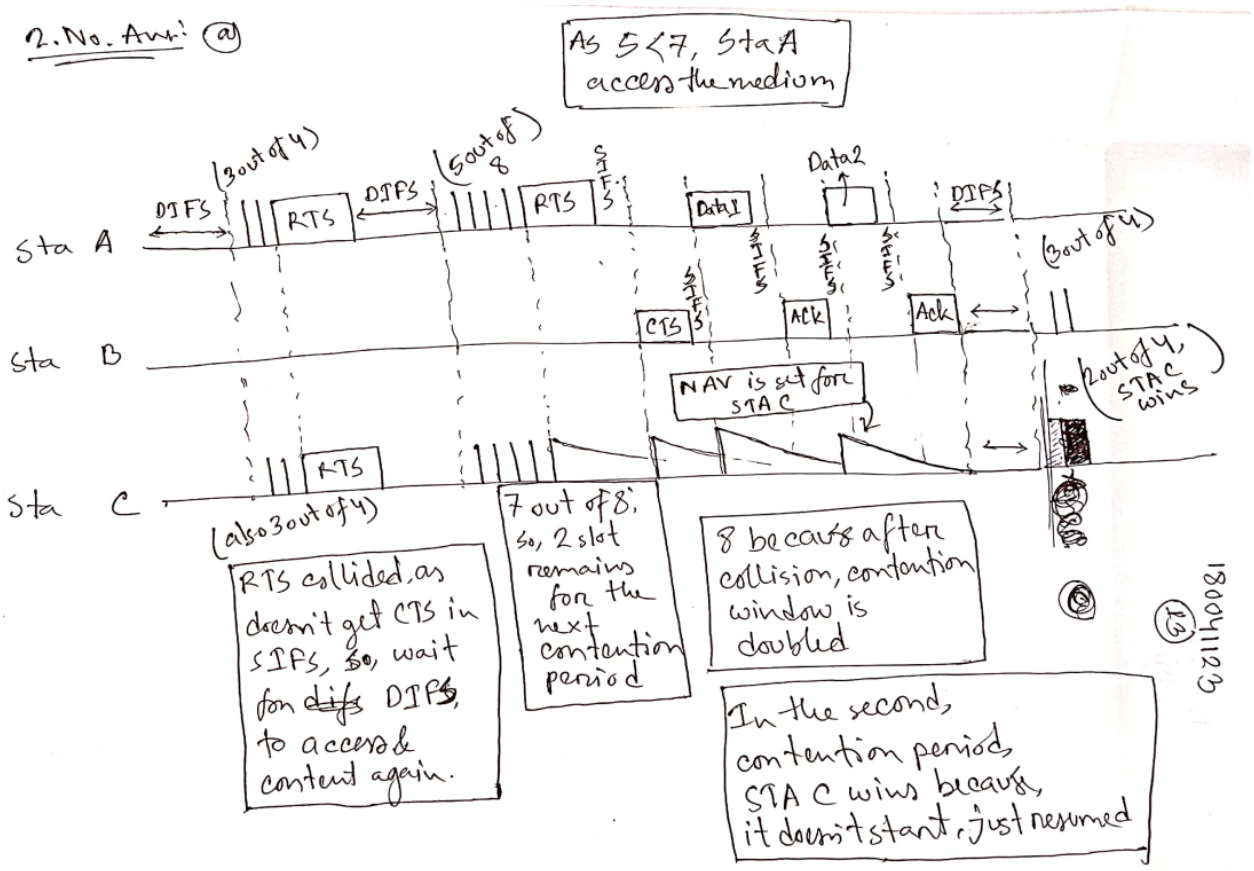
$= 21.3$ msec.

(Ans.)

2. a) Consider a *Basic Service Set (BSS)* of IEEE 802.11 wireless LAN (WLAN) consists of three stations *(A, B, and C)*. Assume that all the stations operate in same frequency band and they can all hear each other transmissions. Consider the *RTS/CTS* transmission is enabled in this scenario. 8.5 (CO3) (PO1)

Draw a Timeline Diagram representing the sequence of actions for a successful Retransmission of two MPDUs (fragmented portion of a single MAC Service Data Unit) from *Station A to Station B.*

The diagram should depict the detailed backoff procedure performed by all the stations in the *BSS*. Consider the minimum *Contention Window, CW_{min}* value is four (4). Note that the x-axis of the diagram shows time and y-axis shows the contending stations in *BSS*. An action (i.e., transmission of a frame) is represented by a horizontal line where the line is placed in the same horizontal line of the station with line length representing period.



b) *Distributed Coordination Function (DCF)* is the fundamental *Medium Access Control (MAC)* technique of the IEEE 802.11-based WLAN standard. *DCF* employs a carrier-sense multiple access with collision avoidance *(CSMA/CA)* with binary exponential backoff algorithm. However, the inherent structure of *DCF* results in decreased throughput efficiency.

i. Identify the root cause behind the throughput inefficiency of IEEE 802.11 *DCF*. 5 (CO2) (PO2)

ii. Propose one possible solution to improve the IEEE 802.11 *DCF* throughput efficiency. 4 (CO4) (PO3)

iii. Identify the challenge(s) associated with your proposed solution. 4 (CO1) (PO2)

i)

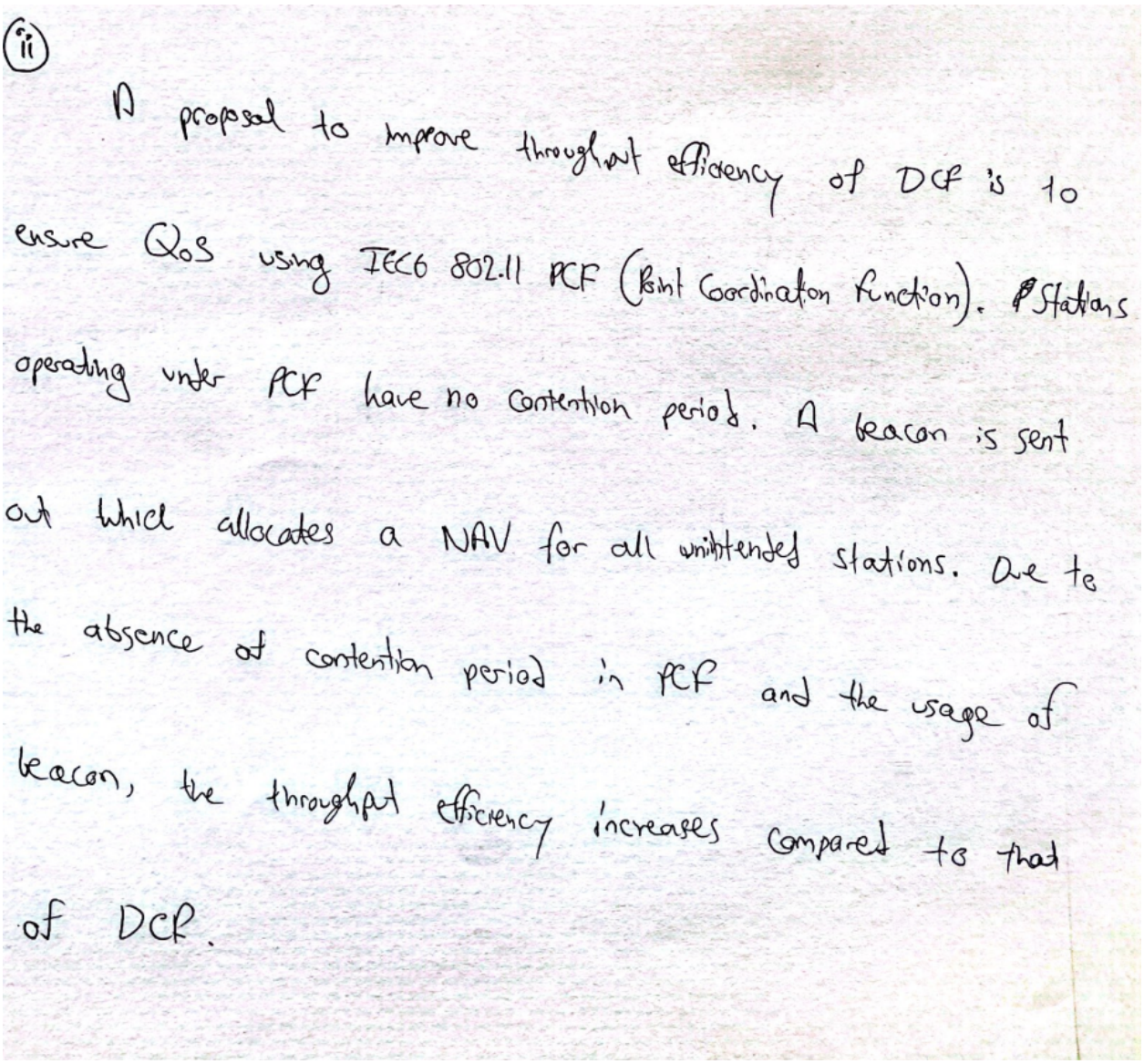
(i)

The root cause of throughput inefficiency of DCF is the fact that there is an exponential backoff period or a contention period. This results is throughput inefficiency as during the contention period the channel sits idle. As a result the data rate decreases of the channel. Also, a station operating under DCF has to wait for a minimum of DIFS time before starting the contention window. All these make the DCF very inefficient in terms of throughput.

There is one paper which states that, DCF causes 30% decrease in the throughput

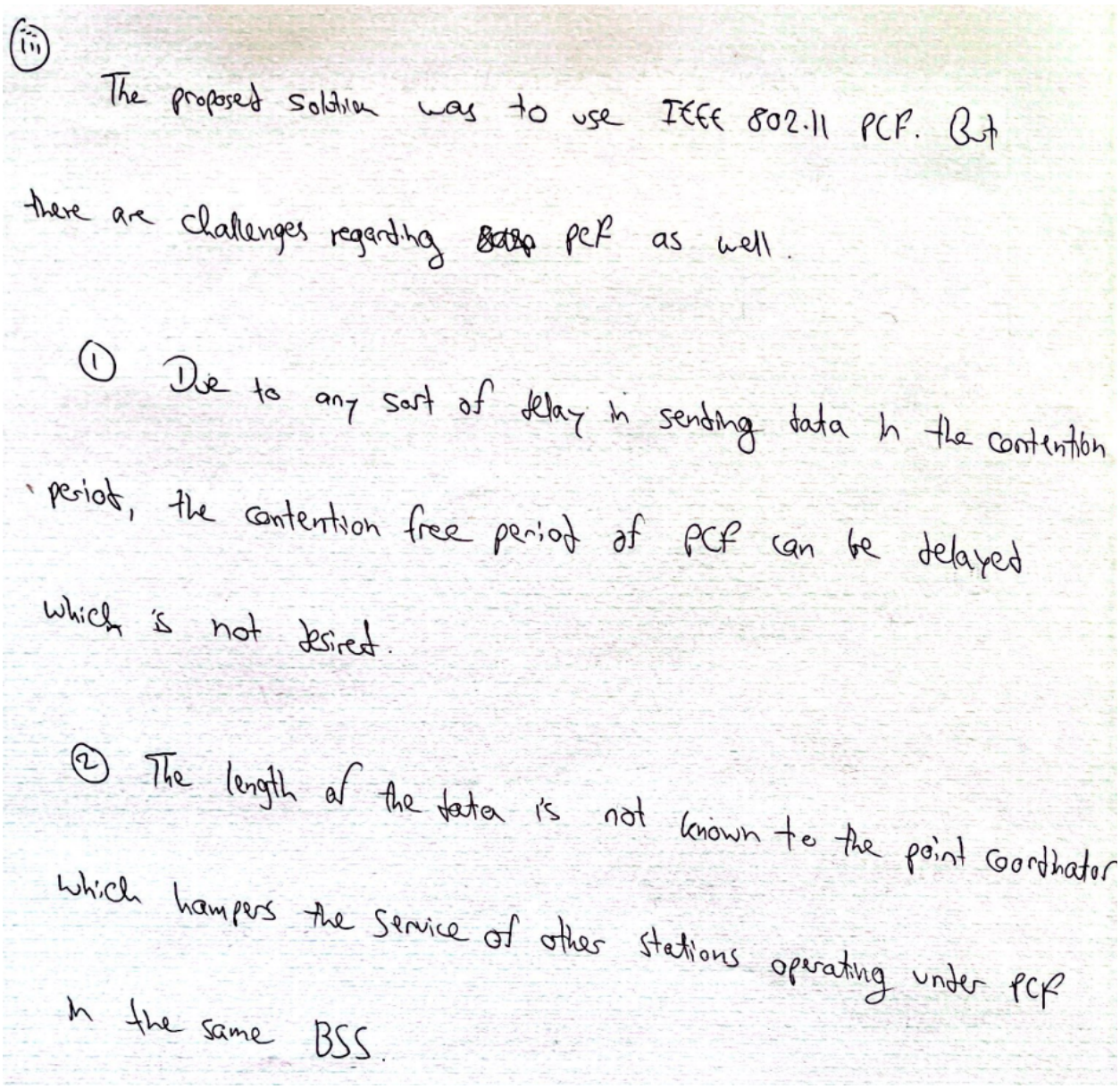
Hence, the root cause for throughput efficiency is backoff & channel stays idle, if two or all station use the same random number.

ii)



This ensures more continuous sending of packets

iii)



1. Complexity: PCF requires a complex coordination mechanism between the access point (AP) and the stations (STAs) in the network. This can increase the overall complexity of the implementation, making it difficult for developers to design and implement a robust and efficient solution.
2. Contention issues: PCF is based on a centralized control mechanism, which can lead to contention issues if multiple STAs attempt to access the network at the same time. This can reduce the overall efficiency of the network and lead to performance issues.

3. Interoperability: There are different implementations of PCF in different 802.11 devices, which can lead to interoperability issues. Ensuring compatibility between different devices can be a challenge for implementers.
4. Limited scalability: PCF is designed for small networks and does not scale well to larger networks with many STAs. This can limit the use of PCF in large enterprise networks or in densely populated areas.
5. Interference: PCF is vulnerable to interference from other devices in the network, which can reduce the overall reliability and stability of the network. Implementers must consider ways to minimize the impact of interference on PCF performance.

3. a) How is the *QoS (Quality of Service)* assured in IEEE 802.11 *PCF (Point Coordination Function)*? Justify your answer with appropriate diagram. 4 (CO1) (PO2)

So, by removing the contention period, PCF assures the QoS.

techniques. Let us see a diagram first.

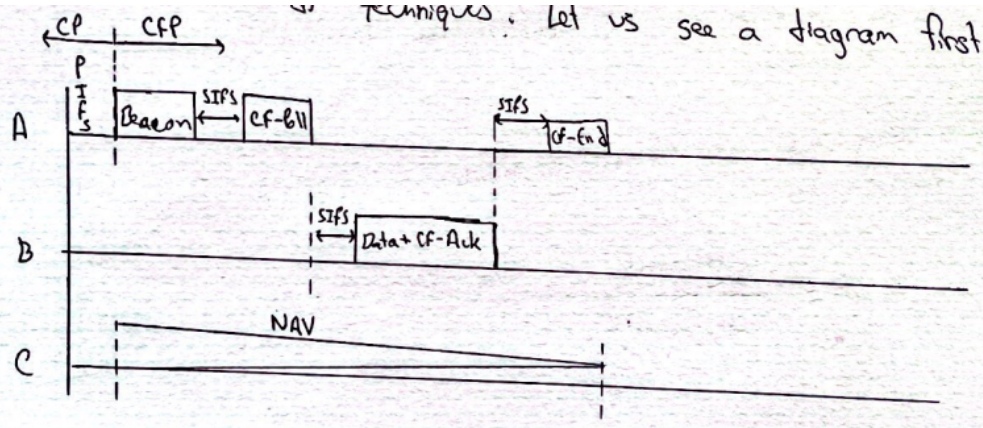


Fig: PCF.

In the diagram be see that station A is operating under PCF. Which means it has to wait PIFS time instead of DIFS time when channel is found idle. Then a beacon is sent out. Note that there is no contention period here which is why it is called a contention free period (CFP). The beacon also holds the NAV timers for all other stations which ensures no collision during this period. The PIFS time is shorter than DIFS time. That combined with no contention period is how QoS is ensured in PCF.

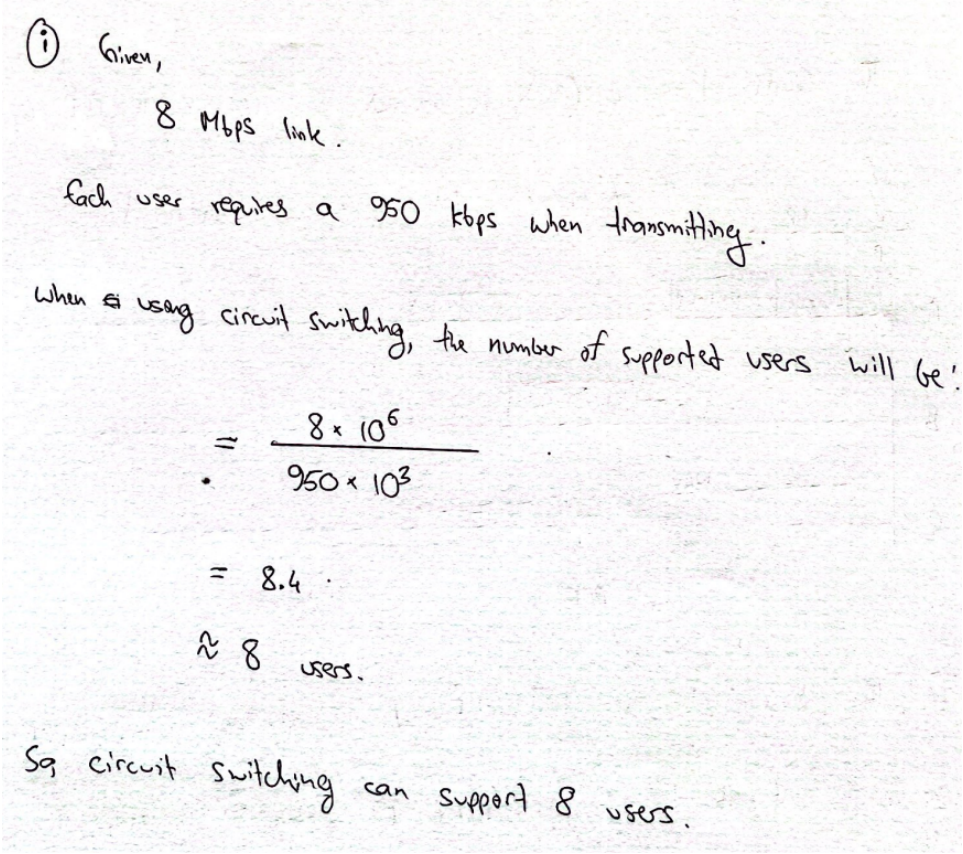
b) Suppose users share a 8 Mbps link. Also suppose each user requires 950 kbps when transmitting, but each user transmits only 18 percent of the time. 5 (CO1)

i. When circuit switching is used, how many users can be supported? (PO2)

ii. For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.

iii. Suppose there are 220 users. Find the probability that at any given time, exactly 90 users are transmitting simultaneously.

i)



ii)

(ii)

It is given that a user transmits 18% of the time.

So, the probability that a given user is transmitting is,

$P(\text{transmit}) = 0.18$

iii)

(iii) Given,

There are 220 users.

We need to find the probability that exactly 90 among the 220 users are transmitting.

This can be done by,

$= {}^{220}C_{90} \cdot (0.18)^{90} \cdot (0.82)^{220-90}$

(Ans).

c) Compare the following terminologies with appropriate examples: 3.5

i. *Multiplexing* (CO1)

ii. *Modulation* (PO2)

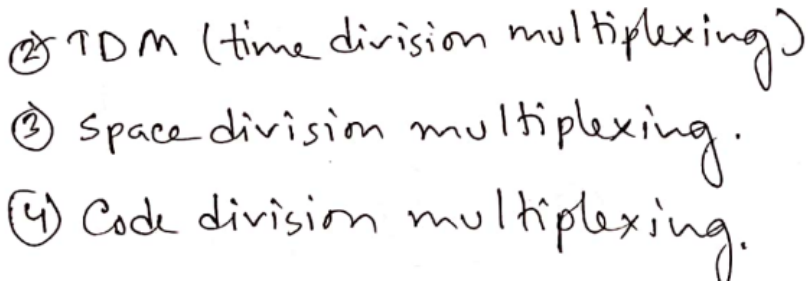
iii. *Multiple Access*

Multiplexing, Modulation and multiple access are three concepts in data communication that are used to enable the efficient transmission of information over a channel.

Multiplexing	Modulation	Multiple Access

Multiplexing	Modulation	Multiple Access
Combining multiple data stream onto a single data stream for transmission over a shared medium by interleaving on time, frequency, code etc	Converting the information signal into a form that can be transmitted over a communication channel. It involves adding a carrier signal to the information signal to create a modulated signal.	Sharing access of a communication channel among multiple devices simultaneously.

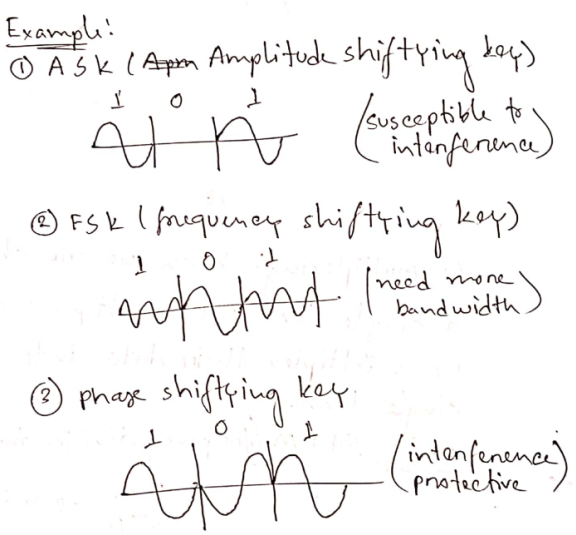
Multiplexing:



Modulation:

Digital Modulation: ASK, FSK, PSK → Digital to analog

Analog Modulation: AM, FM, PM → shifts center frequency of baseband signal up to the radio carrier



The benefits are:

1. Smaller antennas
2. Enables frequency division multiplexing

Multiple Access:

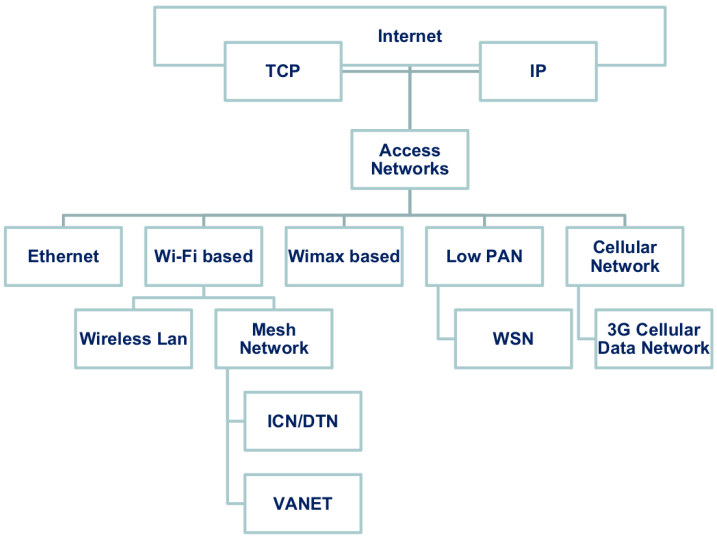
(iii) Multiple Access

FDMA, TDMA, and CDMA are examples of multiple access. In FDMA the channel is divided into multiple frequencies for the stations.

Time is divided among the stations in TDMA. And in CDMA each channel is given a unique code.

d) Define *Access Networks* with appropriate taxonomy. 2.5 (CO1) (PO1)

- The connection between an end device (such as a computer or phone) and the first router (edge router), on its path to other end devices.
- It is the first entry point to internet for the users



Taxonomy of access networks

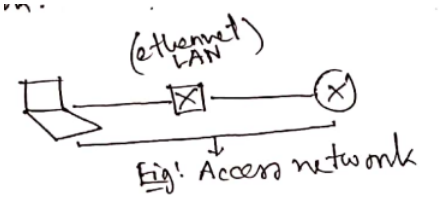


Fig: Access network

Example: ① Wifi
② 3G or 4G network
③ DSL (digital subscriber line)
④ HFC (Hybrid fiber Co-axial)
⑤ FTTH (Fiber To The Home)